Problem 1

Let $\mathcal{F}$ and $\mathcal{G}$ be σ-fields of subsets of Ω.

(a) Use elementary set operations to show that $\mathcal{F}$ is closed under countable intersections; that is, if $A_1, A_2, \ldots$ are in $\mathcal{F}$, then so is $\bigcap_i A_i$.

(b) Let $\mathcal{H} = \mathcal{F} \cap \mathcal{G}$ be the collection of subsets of Ω lying in both $\mathcal{F}$ and $\mathcal{G}$. Show that $\mathcal{H}$ is a σ-field.

(c) Show that $\mathcal{F} \cup \mathcal{G}$, the collection of subsets of Ω lying in either $\mathcal{F}$ or $\mathcal{G}$, is not necessarily a σ-field.

Proof.

(a) :Let $A_1, A_2, \ldots \in \mathcal{F}$. Then $A_1^c, A_2^c, \ldots \in \mathcal{F}$.

Since $\mathcal{F}$ is a σ-field, we have $\bigcup_i A_i \in \mathcal{F}$.

By De Morgan's Law, $\left(\bigcap_i A_i\right)^c = \bigcup_i A_i^c \in \mathcal{F}$.

Hence, $\bigcap_i A_i \in \mathcal{F}$.

(b) :1. Since $\Omega \in \mathcal{F}$ and $\Omega \in \mathcal{G},$ we have $\Omega \in \mathcal{H}$.

2. $\forall A \in \mathcal{H},$ we have $A \in \mathcal{F}$ and $A \in \mathcal{G},$ so $A^c \in \mathcal{F}$ and $A^c \in \mathcal{G},$ hence $A^c \in \mathcal{H}$.

3. $\forall A_1, A_2, \ldots \in \mathcal{H},$ we have $A_1, A_2, \ldots \in \mathcal{F}$ and $A_1, A_2, \ldots \in \mathcal{G}$,

so $\bigcup_i A_i \in \mathcal{F}$ and $\bigcup_i A_i \in \mathcal{G},$ hence $\bigcup_i A_i \in \mathcal{H}$.

(c) :Counterexample: Let $\Omega = \{1, 2, 3\}, \mathcal{F} = \{\emptyset, \{1, 2\}, \{3\}, \Omega\}, \mathcal{G} = \{\emptyset, \{1\}, \{2, 3\}, \Omega\}$.

Then $\{1, 2\} \in \mathcal{F}$ and $\{1\} \in \mathcal{G},$ but $\{1, 2\} \cup \{1\} = \{1, 2\} \notin \mathcal{F} \cup \mathcal{G}$.

Hence, $\mathcal{F} \cup \mathcal{G}$ is not necessarily a σ-field.

□

Problem 2

Let $A_1, A_2, \ldots$ be a sequence of events. Define

$$B_n = \bigcup_{m=n}^{\infty} A_m, \quad C_n = \bigcap_{m=n}^{\infty} A_m$$

Clearly $C_n \subseteq A_n \subseteq B_n$. The sequences $\{B_n\}$ and $\{C_n\}$ are decreasing and increasing respectively with limits

$$\lim B_n = B = \bigcap_n B_n = \bigcap_n \bigcup_{m \geq n} A_m, \quad \lim C_n = C = \bigcup_n C_n = \bigcup_n \bigcap_{m \geq n} A_m$$

The events B and C are denoted $\limsup_{n\to\infty} A_n$ and $\liminf_{n\to\infty} A_n$ respectively. Show that

(a) $B = \{\omega \in \Omega : \omega \in A_n$ for infinitely many values of $n\}$,

(b) $C = \{\omega \in \Omega : \omega \in A_n$ for all but finitely many values of $n\}$,

We say that the sequence $\{A_n\}$ converges to $A = \lim A_n$ if B and C are the same set A.

Suppose that $A_n \to A$ and show that

(c) A is an event, i.e. $A \in \mathcal{F}$,

(d) $\mathbb{P}(A_n) \to \mathbb{P}(A)$.

Proof.

(a) :Since by defination: $B = \bigcap_n \bigcup_{m \geq n} A_m$

$\Rightarrow$Suppose $\omega \in B$, then $\omega \in \bigcup_{m \geq n} A_m$ for all $n \in \mathbb{N}$

This means for each n, we can find an $m \geq n$ such that $\omega \in A_m$

Hence, $\omega \in A_n$ for infinitely many values of n

$\Leftarrow$Suppose $\omega \in A_n$ for infinitely many values of n,

then for each n, we can find an $m \geq n$ such that $\omega \in A_m$

This means $\omega \in \bigcup_{m \geq n} A_m$ for all $n \in \mathbb{N}$

Hence, $\omega \in \bigcap_n \bigcup_{m \geq n} A_m = B$

(b) :$\Rightarrow$Suppose $\omega \in C$, then $\exists n \in \mathbb{N}$ such that $\omega \in \bigcap_{m \geq n} A_m$

This means $\exists n \in \mathbb{N}$ such that $\omega \in A_m$ for all $m \geq n$

Hence, $\omega \in A_n$ for all but finitely many values of n

$\Leftarrow$Suppose $\omega \in A_n$ for all but finitely many values of n,

then $\exists n \in \mathbb{N}$ such that $\omega \in A_m$ for all $m \geq n$

This means $\omega \in \bigcap_{m \geq n} A_m$

Hence, $\omega \in \bigcup_n \bigcap_{m \geq n} A_m = C$

(c) :Since $A_n \to A,$ so $\lim\limits_{n\to+\infty} A_n = \limsup\limits_{n\to+\infty} A_n = \liminf\limits_{n\to+\infty} A_n$

So we have $A = B = C$

Since B_n are countable unions of $A_m,$ so $B_n \in \mathcal{F}$ for all $n \in \mathbb{N}$

Then, $B = \bigcap\limits_n B_n \in \mathcal{F}$

Hence, we have $A \in \mathcal{F}$

(d) :Since $C_n \subseteq A_n \subseteq B_n,$ we have $\mathbb{P}(C_n) \le \mathbb{P}(A_n) \le \mathbb{P}(B_n)$

Since C_n increases to C, B_n decreases to B,

we have $\mathbb{P}(C_n) \to \mathbb{P}(C) = \mathbb{P}(A), \mathbb{P}(B_n) \to \mathbb{P}(B) = \mathbb{P}(A)$

Hence by squeeze theorem, $\mathbb{P}(A_n) \to \mathbb{P}(A)$

$\square$

Problem 3

Let $\mathcal{E}$ be the left open right closed intervals of $\Omega := \mathbb{R}$ defined in class. Write down the algebra generated by $\mathcal{E}$, and prove your result.

Problem 4

Let Ω be a sample space. Show that any finite algebra on Ω is a $\sigma-$algebra.